Lane Perception for Automated Driving with Human-Anchored, Engine-Certified Ground Truth: An Explicit Counting Contract and Pre-Registered Single-Factor Evaluation

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Abstract

Lane perception for automated driving is bottlenecked by two conflated problems: the cost of pixel-accurate annotation, and the absence of an agreed definition of a "lane line". We present a closed-loop research programme that attacks both. Every label carries an explicit provenance credential --- human revision, agent review, or engine-certified truth --- so no claim has to guess what truth it rests on, and the reported rounds added no manual annotation. Evaluation is made explicit and versioned: a *counting contract* gives every ratio an explicit numerator and denominator, and a *protocol* fixes the geometric scope of a line candidate, with a hash binding the sealed set to its definition. One segmentation arm (six seeds) is evaluated on two reference sets, a road-disjoint held-out set and in closed-loop driving. Three findings follow. (i) With the weights fixed, changing the post-processing switches and the evaluation scope changes the readings and can reverse the ranking of two arms; about 30% of annotated line pixels fail the appearance predicate the adopted definition uses. (ii) The one-shot confirmation does **not** reproduce the acceptance numbers (label-scope recall 0.460, precision 0.763) --- the headline of that independent test. (iii) The same definition change came with no observed centre-line crossing in four runs, a 66% drop in lane-source availability, and an acceptance gate still failing 0/4; the blocker is a measured static body-sweep deadlock. All changes were pre-registered and judged by rules fixed before the runs.

Index Terms—Lane detection, ground-truth generation, evaluation protocol, automated driving,

Highlights

- Ground-truth provenance is credentialed and enforced: human revision, agent review, engine-certified.
- A counting contract gives every reported ratio an explicit numerator and denominator.
- The protocol hash binds a sealed held-out set to the definition it is read under.
- The two natural definitions of a lane line fail different gates on identical data.
- A one-shot held-out confirmation does not reproduce the development-set numbers.
- Five pre-registered driving factors were rejected; the blocker is a measured deadlock.

I. Introduction

Lane-keeping is the lowest-level competence of an automated driving stack, and it is still surprisingly hard to evaluate. Two obstacles recur in the literature. The first is annotation cost: pixel-accurate lane labels are expensive, so public datasets are small relative to the diversity of road appearance, and every new operational domain demands a new labelling campaign. The second is definitional: papers report "lane detection" accuracy against ground truth that may include or exclude faded, occluded, dashed, yellow, or shadowed markings, and may or may not count markings beyond the ego lane. Two systems can therefore disagree by tens of accuracy points while disagreeing about nothing except the definition of the object they detect.

This paper reports a research programme built around those two obstacles. It is not a new network architecture; it is an attempt to make the *data* and the *evaluation* rigorous enough that architectural progress can be measured. The programme has three components.

Credentialed truth provenance. Every label in this programme carries a provenance credential, and the credential is enforced rather than declared: a command-line claim cannot promote a label (Section III-B). The ladder has three rungs -- human frame-by-frame revision, per-frame agent review, and engine-derived truth -- and the engine rung is admitted only through four explicit certification gates: a credential gate (the label source must be declared and verifiable), a paint-appearance gate (the pixels claimed as line must look like paint in the image), a road-type gate (the frame must carry the road-type column that makes pavement decidable), and a negative-evidence gate (frames claimed to contain no line must show evidence of their own, not merely the absence of a label). The rounds reported here add no new manual annotation: the experimental variable is the composition of the engine-certified increment on top of the human-revised anchor, and Section VII states what the anchor's small size costs us.

An explicit counting contract. Every reported ratio is defined by a counting contract that names its numerator and denominator

over an explicit set chain: truth-confirmed paint frames, evaluated perception candidates on those frames, candidates with a reference on their own side, matched candidates, matches with decidable predicted and reference roles, and role-agreeing matches. Coverage, identity and role are therefore instance-level ratios with stated denominators; precision and recall remain pixel-level and are reported in two *scopes* (label pixels, and paint-like pixels) because the two answer different questions.

Versioned protocol definitions with hash-gated evaluation. A protocol fixes the geometric scope of a line candidate and is a first-class, versioned artefact. A protocol change alters the evaluation hash, and both the frozen held-out set and the one-shot confirmation record that hash, so a confirmation produced under one definition cannot be silently attributed to another. This mechanism caught a real defect during the work: before it was fixed, the hash did not follow the protocol, and a set sealed under the old definition would have authorised a confirmation under the new one.

Method discipline. Every change in this paper was pre-registered with its reading rule before the runs, and each pre-registration names the protocol version and the repository commit it was frozen against; the rules include safety non-inferiority read as a maximum over runs, an explicit refusal to treat an unmeasured quantity as a pass, and a prohibition on relaxing the frozen gates. Changes were applied one factor at a time and judged against alternating control/factor runs. Where a pre-registered rule produced a verdict we considered too coarse, we report the verdict as given and record the rule's defect rather than reinterpreting the result; one such defect — safety metrics that are frame *counts* and therefore scale with how far the car travelled — is analysed in Section V-F and quantified in the supplement.

Contributions. (1) A credentialed truth-provenance ladder (human revision / agent review / engine-certified) with enforcement, plus a certification-gated engine pipeline for the machine-generated increment; the composition of every pool is stated explicitly, and the reported rounds add no new manual annotation. (2) The counting contract, and the demonstration that the two natural readings of "line" fail different gates on the same data — a definitional disagreement that moves about 30% of annotated line pixels out of the adopted scope (which retains 69% of annotated line pixels on the development pool and 89% on the limited-class pool). (3) A protocol/version/hash mechanism that binds a held-out set to the definition under which it is read. (4) A one-shot, road-disjoint confirmation with an explicit UNKNOWN policy, whose negative result (no reproduction of the development-set numbers) we report as the finding. (5) A measured localisation of why a perception stack that passes its offline gates still fails a closed-loop acceptance gate, including a static-deadlock mechanism and the observation that the "on-pavement" gate was in fact being driven by a neural obstacle layer rather than by pavement evidence.

II. Related Work

Lane detection methods. Modern lane detection is dominated by deep segmentation and by parametric or anchor-based curve regression. Encoder–decoder segmentation with dilated convolutions and a discriminative loss (LaneNet [1]) and message-passing across rows (SCNN [2]) established the segmentation line of work that the present system follows; cross-layer refinement with anchor-based heads (CLRNet [3]) improved curve accuracy and speed. Chinese-language work in the same family explores lightweight variants of CLRNet [4], B-spline end-to-end regression [5], graph/transformer hybrids [6], transformer with cross-attention [7], parallel detection over image sequences [8], multi-scale feature aggregation [9], DeepLabV3+ variants [10] of the original architecture [11], symmetry-prior and key-point methods for complex scenes [12], and 3-D lane estimation with convolutional architectures [13]. These works share our object of study but differ in what they optimise: they report accuracy against datasets whose annotation conventions differ from each other, which is exactly the ambiguity this paper makes explicit.

Datasets and ground truth. TuSimple [14], CULane [2] and Cityscapes [15] remain the standard benchmarks; their annotation conventions (lane presence per row, presence of far-field markings, treatment of occlusion) differ, and neither reports the annotation's *appearance* properties. Ground-truth generation has been studied as a problem in its own right: Borkar et al. [16] built an automated ground-truth process around time-slicing visualisation for a night-time lane detector, and McCall and Trivedi [17] surveyed and evaluated video-based lane estimation with an explicit eye on evaluation methodology. Our work continues that line with two additions: truth derived from engine state rather than from imagery, and a counting contract that makes the denominators of every reported ratio inspectable.

Automatic labelling and simulation. Self-training and auto-labelling are widely used to reduce annotation cost, but their evaluation inherits the definitional problem above. Simulator-derived labels are common in synthetic benchmarks; less common is the discipline of certifying each generated label against image evidence, which we adopt here because a label that cannot be recovered from the image is not a learnable target.

Evaluation discipline. Pre-registration is standard in empirical sciences and rare in perception research. Our contribution here is not the concept but a worked, fully recorded instance: a pre-registration, an alternating single-factor A/B, a fixed reading rule, and a verdict per factor — including five factors whose verdict was negative.

III. System and Method

A. Overall architecture

The stack is a perception-led driving system. A camera ring (eight mounts) and a LiDAR range image feed a semantic segmentation head whose line output is converted to world-space markings; a lane reference is selected from those markings under the protocol definition; a planner produces a path; a safety monitor arbitrates the tick against hard checks (pose, boundaries, occupancy, obstacle risk) and publishes a speed and level; the controllers consume the arbitrated path. The iron rule of the stack is that the lateral reference comes from perception only — the map answers "where to", never "where the lane is". the supplement shows the candidate-scope pipeline with the measured revocation rate discussed in Section V-E.

B. Truth provenance, credentials, and certification gates

Two dimensions must not be conflated here. **Provenance** says who or what produced a label (human revision, agent review, engine); **certification** says whether that label passed the four gates below. A human-revised frame can fail the appearance gate and an engine frame can pass all four, so admission is decided by the gate result, never by the rung. With that separation stated, the programme distinguishes three rungs of truth and records the rung with every label: **human revision** (a human annotates or corrects the line labels frame by frame), **agent revision** (a per-frame agent review of engine output), and **engine-certified** truth (the engine's own line geometry projected into the camera). Two pools are human-revised: the training anchor (5 directories, 75 frames) and the development/evaluation packs (9 packages, 117 frames), which is why the acceptance results of Section V-A are measured against human annotation. The machine-generated increment that the ablations vary is engine-certified; it is admitted only when: (i) the credential is present and parseable, and the label source is declared (frames whose provenance cannot be named are excluded rather than guessed); (ii) the pixels claimed as line satisfy a paint-appearance predicate — white paint requires high value, low chroma and a red/blue balance; yellow paint requires a hue ratio band; the predicate is a single-source module so that a threshold change is a versioned change, not a local edit; (iii) the frame carries the road-type column, without which "on pavement" is undecidable; (iv) a frame claimed to contain no line carries positive evidence of its own (a certified negative), so that "no label" is never read as "no line". Frames failing any gate are excluded from both training and evaluation and are counted in the audit, not silently dropped.

The gate bookkeeping is itself a result: the supplement shows the negative-pool funnel for one collection round (admitted, suspect, contains paint, undecided) — a pool that looked plentiful on disk shrank to a certified subset before it was ever used as training signal.

C. The counting contract

The contract is stated over integer counters that are accumulated per frame; the ratios are computed once from those integers, never by averaging per-directory ratios. P counts frames whose annotation carries a line on pavement (decided frame by frame: a frame is not admitted because another frame in its group has a line); C counts the perception candidates inside P frames that enter the evaluation; C_{out} counts candidates on frames outside P and is reported separately; $R \subseteq C$ counts candidates whose own side has an available reference; $M \subseteq R$ those satisfying the frozen matching condition; $L \subseteq M$ those where both the prediction and the reference have a decidable left/right role; and $A \subseteq L$ those whose role agrees. The reported instance-level ratios are coverage $|R|/|C|$, identity $|M|/|R|$ and role agreement $|A|/|L|$; pixel-level precision and recall are reported in two scopes (label, paint). 1 shows the contract.

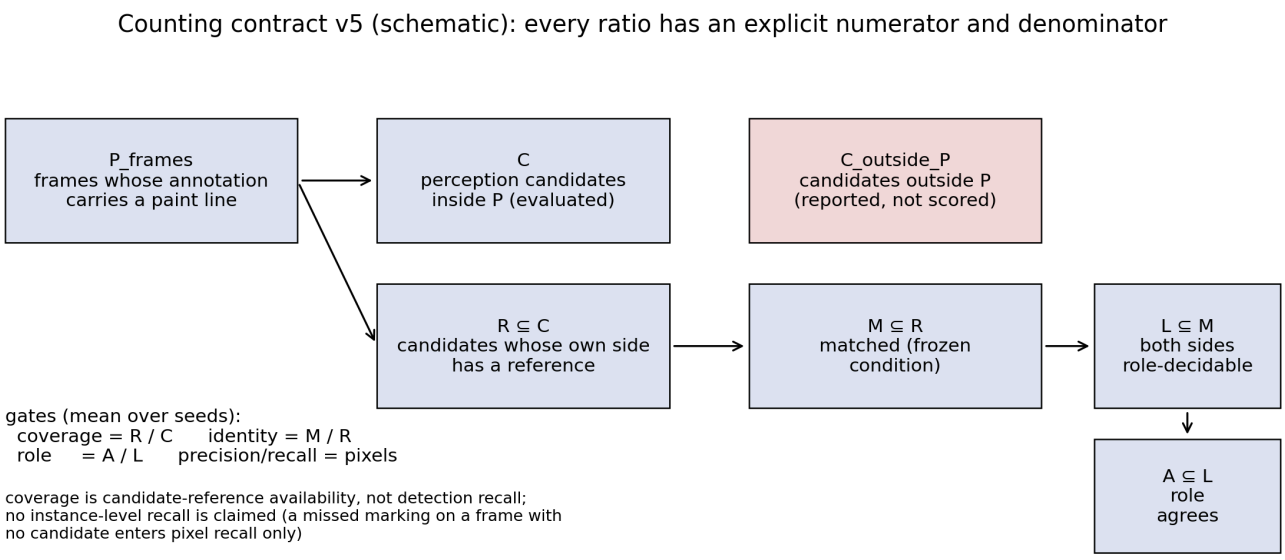


Fig. 1. Counting contract (schematic): every ratio has an explicit numerator and denominator, over the frozen counter definitions; coverage is candidate-reference availability, not detection recall.

Two consequences have to be read off this definition rather than assumed. First, coverage is the **candidate-reference availability**: the share of evaluated candidates whose own side has truth to compare against, not the share of true markings that were detected. A line-bearing frame on which the perception produces no candidate contributes nothing to $|R|/|C|$; its miss is carried by pixel-level recall instead. Second, candidates on frames outside P are counted as C_{out} and reported, so that "no reference available" is never scored as a coverage failure. We therefore claim no instance-level recall: the three conditional ratios above cannot substitute for one, and reporting one would require the total number of ground-truth instances together with an explicit one-to-one assignment rule. Every ratio still has an explicit denominator, and the frame-level counter P is reported separately from the candidate-level counters, so a run with few candidate-bearing frames cannot masquerade as a good one.

D. Protocol definitions

A protocol fixes three things about a line candidate: its **lateral scope** (candidates farther than a threshold from the ego's own lane are not "my lane's lines"), whether **same-side near-duplicate markings are merged** into one instance, and whether an **appearance gate** removes candidates whose pixels do not look like paint. The frozen baseline protocol applies none of the three (lateral scope off, merging off, appearance off); the adopted protocol applies all three with a lateral threshold of 5.5 m, a merge radius of 1.0 m and the appearance predicate of Section III-B. the supplement shows the switch matrix and the live hash; the hash is a function of the active protocol, so evaluation artefacts cannot be reattributed across definitions.

The lateral scope is not arbitrary: it is set from the measured geometry of the reference pool, shown in the supplement (261 042 marking pixels, median offset −0.17 m, median absolute offset 1.22 m). A scope of 5.5 m retains the own lane's markings and their immediate neighbours while excluding the far-field markings that dominate the false pairings discussed in Section V-E.

E. Training-composition design space

Because ground truth is generated rather than annotated, the composition of the training pool is a first-class experimental variable. We study it along one axis: the **negative dose**, i.e. how many certified line-free frames are mixed into the pool relative to the base composition, together with composition variants that place markings in specific geometric relations (near/far pairs, same-side pairs). Section V-C reports the resulting dose–response.

F. Held-out final set: sealing, hash gating, one-shot consumption

An independent confirmation set is composed under three constraints: it must be **road-disjoint** from every training pool (identity taken from the generation metadata, not from directory names), it must be certified by the gates of Section III-B, and its content must be frozen by a digest over per-frame hashes. Sealing records the frame digests, the protocol hash and the composition; access is then gated on the protocol hash, and a single confirmation **consumes** the set: after a successful confirmation the set may not be read again, and a refused access is recorded in a ledger without consuming it. The confirmation program evaluates the candidate on the sealed frames and records which quantities were measured; quantities that were not measured are recorded as UNKNOWN and may not be reported as confirmed. The full flow, with the live seal, is shown in the supplement.

G. Driving stack and safety arbitration

In closed loop the same perception feeds a planner and a safety monitor whose arbitration ladder evaluates rules in a fixed order; soft rules accumulate speed caps, while a rule whose worst level is `minimal_risk` stops the tick. the supplement lists the ladder. Two rules matter for Section V-E: the **planned body-sweep rule**, which stops the tick when the body sweep of the planned path would cross a lane boundary closer than a hard threshold (4 m), and the **on-pavement gate**, which revokes the perception lane when the lane centre does not lie on observed drivable surface.

IV. Experimental Setup

Reference sets. Two disjoint sets are used for the primary acceptance: a development pool of 9 certified scenes (117 frames) and a limited-class pool of certified scenes chosen for near-field background structure (24 frames per arm). Neither shares frames with training pools.

The limited-class pool is selected, not sampled: candidate scenes are ranked by a model-independent near-field structure-density criterion computed from the human annotation (the supplement), so the pool concentrates on the scenes where identity and role are hardest rather than on the average scene. A third, road-disjoint set of 200 frames (96 line-bearing) is sealed for the one-shot confirmation of Section V-D.

Disjointness is audited with pose, not with names: every candidate collection is checked against the development anchors under a 50 m buffer and gets a recorded verdict (supplement). The same machinery rejected the first two compositions of the sealed set, which is why the sealed composition also carries a content-level audit (the supplement).

Arms. All segmentation arms share one initialisation, one budget and one recipe; they differ only in training-pool composition (the dose arms) or in a single post-processing factor (protocol components). Composition is explicit: a human-revised anchor of 5 directories (75 frames) plus a varying number of engine-certified packages (2 paired-line packages of 16 frames and 12 certified negative packages of 96 frames in the reported base), with the negative dose as the varied axis. Six seeds are used for acceptance; composition arms are reported with their seed counts.

Metrics and gates. The frozen gates are coverage $\geq$ *0.80*, identity $\geq$ *0.60*, precision $\geq$ *0.40*, recall $\geq$ *0.70* and role agreement $\geq$ *0.70*; recall is read in the paint scope for the adopted protocol and in the label scope for the baseline, and both readings are always reported. The verdict rule changed between the two rounds and the change is part of the record rather than a detail: the baseline round's stored verdicts require **every seed** to pass, while the adopted round stores `verdict_mode = mean` and reports the per-seed values alongside. Section V-A therefore gives the stored verdicts and a re-tabulation of the same per-seed numbers under each rule, so that no gate outcome has to be read under an unstated rule.

Driving experiments. Closed-loop runs use one fixed scenario (a town link), alternating control and factor arms, four runs per arm; hard targets are the benchmark's fixed checks (frames, reversing, centre and edge crossings by reference line *and* by body sweep, stall, on-road, collisions, goal). Reading rules were fixed before each round: hard gates require 4/4, safety items are compared as the maximum over runs, availability as median with range, and an UNKNOWN never releases a gate.

Discipline. Every factor in Section V was pre-registered with its reading rule; no threshold was relaxed at any point;

Timing hygiene is recorded rather than estimated: the supplement is the repeated-measurement protocol used before any latency claim (quiet-machine precondition, repeated runs, smaller value taken — a leaked simulator instance once inflated p95 by a factor of 7.6). every verdict file, manifest and per-frame telemetry trace is retained, and every figure in this paper is regenerated from those files by two scripts.

V. Results

A. Acceptance under two definitions

Table I reports the two definitions on both sets, and Table II re-tabulates the recall gate of the same checkpoints under both post-processing settings and both scopes. Under the baseline definition the stored verdict fails identity on **both** sets (0.433 dev, 0.414 limited-class) and additionally precision (0.402) and role (0.674) on the limited-class set; under the adopted definition all five gates pass on both sets by the mean rule (coverage 0.956/0.861, identity 0.665/0.716, precision 0.894/0.659, paint recall 0.812/0.885, role 0.883/0.793). 2 shows the same numbers against the thresholds.

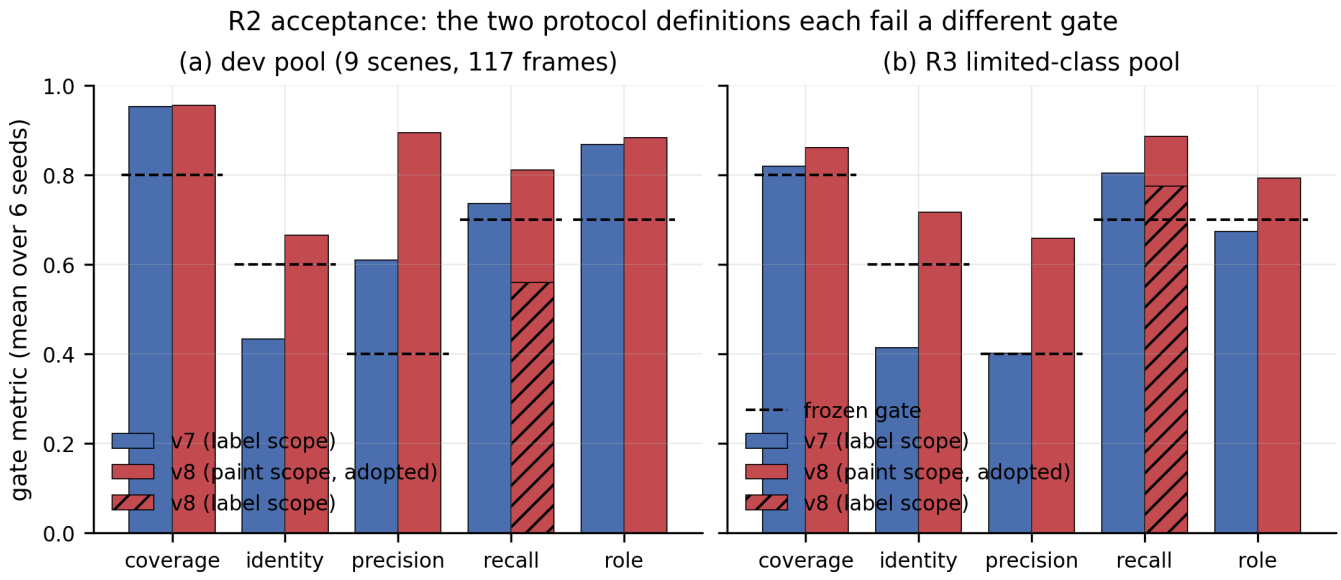


Fig. 2. Gate metrics against the frozen thresholds for the two protocol definitions, development pool (a) and limited-class pool (b).

Three qualifications belong with this table. (i) The two rows were not judged by the same rule. Re-tabulated under a common mean rule, the baseline's limited-class precision (0.4018) is above its 0.40 threshold even though 4 of 6 seeds fall below it, and the baseline's development recall (0.736) is above its 0.70 threshold even though 2 of 6 seeds fall below it; under a common all-seeds rule the adopted definition's development label-scope recall fails 6 of 6 seeds while its paint-scope recall fails 1 of 6. (ii) The protocol change moves three post-processing switches and the evaluation scope at the same time, so this table cannot attribute the whole difference to the definition; Table II separates the scope from the post-processing for recall, and the per-switch single-factor results are in Section V-B. (iii) The candidate-level gates are the stable part of the picture: the identity gate is missed by 6 of 6 baseline seeds on both sets and role by 5 of 6 on the limited-class set, while under the adopted definition no seed misses any of the three (the supplement).

the supplement plots the limited-class pool seed by seed under both definitions, with each candidate-level gate marked when missed; it is drawn from the same two verdict files as Table I, so the figure and the table cannot disagree about which run produced them.

Table I. The two protocol definitions on both reference sets, as stored by each round under its own verdict rule (means over six seeds; bold marks a gate the stored verdict failed; the adopted protocol is read in paint scope, the baseline in label scope). The rule differs between the two rows — see the text and Table II.

Set	Definition	Coverage	Identity	Precision	Recall	Role
Dev (9 scenes, 117 frames)	baseline (label scope)	0.952	**0.433**	0.610	0.736	0.868
Dev	adopted (paint scope)	0.956	0.665	0.894	0.812	0.883
Limited-class (24 frames)	baseline (label scope)	0.819	**0.414**	**0.402**	0.804	**0.674**
Limited-class	adopted (paint scope)	0.861	0.716	0.659	0.885	0.793

Table II. The recall gate re-tabulated: the same six checkpoints per pool, read under both post-processing settings and both scopes, with the number of seeds below the 0.70 threshold. The paint scope reads each checkpoint under the adopted definition; the label scope under the baseline one.

Pool	Post-processing	Scope	Mean recall	Seeds below 0.70
Development	baseline switches (v7)	label	0.736	2/6
Development	baseline switches (v7)	paint	0.813	1/6
Development	adopted switches (v8)	label	0.559	6/6
Development	adopted switches (v8)	paint	0.812	1/6
Limited-class	baseline switches (v7)	label	0.804	0/6
Limited-class	baseline switches (v7)	paint	0.891	0/6
Limited-class	adopted switches (v8)	label	0.775	0/6
Limited-class	adopted switches (v8)	paint	0.885	0/6

B. The disagreement is definitional

Two things move together here and they have to be separated. The **scope** changes the reading: Table II shows that under a fixed post-processing setting, moving from the label scope to the paint scope changes development recall from 0.736 to 0.813 (baseline switches) and from 0.559 to 0.812 (adopted switches) — in the second case a 25-point swing with no change to the weights. The **post-processing** also changes the reading, and not in the same direction: with the scope held at label, the adopted switches cost 18 points of recall (0.736 → 0.559). Table V, which turns each switch on alone, attributes that cost entirely to the appearance gate: the lateral scope and the merge leave label recall untouched (0.736 in both) and act on identity instead (0.432 → 0.621 and 0.432 → 0.459 respectively), while the appearance gate is the only switch that trades label recall for pixel precision. What the scope change is worth in pixels is measured, not assumed: the paint scope retains 69% of annotated line pixels on the development pool and 89% on the limited-class pool, and the supplement shows the per-seed label-scope and paint-scope recall with the non-paint fraction of label line pixels overlaid (~0.30). Those excluded pixels are the ones the appearance predicate does not select; Section VI states what that does and does not license us to claim.

The instance-level gate is more sensitive than that, and in a way that matters for model selection. the supplement reads 20 checkpoints (two arms, ten seeds each) under the label-scope and surface-scope identity definitions: the base arm moves from a mean of 0.399 to 0.867 and the 6× arm from 0.429 to 0.771. The absolute values nearly double for one arm and fall for the other, and **the ranking of the two arms reverses** (base below 6× by 0.031 under the label scope, above it by 0.095 under the surface scope). The same reversal is recorded in the repository's protocol proposal for the surface scope, which is why that scope is reported rather than promoted: a gate that can change which checkpoint is selected is a model-selection decision, not a reporting detail. An identity threshold must therefore name its scope, exactly as the recall gate does. This experiment uses its own pool and its own ten seeds per arm, so its numbers are not comparable with Table I's six-seed means and are not presented as such.

The effect is not specific to the delivered arm. On a second arm trained on a different composition (near/far pair packages, six seeds, same frozen protocol), moving from the label to the paint scope raises recall from 0.459 to 0.673 (+0.214) against +0.252 for the delivered arm, and the identity gate reads 0.658 against 0.665. The direction of the scope effect reproduces on the second model; its magnitude is arm-specific, which is why this paper reports the scope with every number instead of quoting a single calibration.

Two further single-factor results constrain how the adopted definition behaves. the supplement (boundary map) sweeps mask dilation: dilation buys recall in both sets but costs precision, and the limited-class set pays much more for it (at one pixel of dilation its precision falls from 0.65 to 0.39, a drop of 0.26, against 0.89 to 0.84, a drop of 0.05, on the development pool; the two rows are read at their own dilation settings), so no single dilation value clears both sets. 3 shows the appearance gate acting exactly as designed: precision rises sharply (dev 0.650 → 0.884; limited-class 0.387 → 0.650) and IoU improves on the limited-class set (0.347 → 0.532) at the cost of recall (dev 0.766 → 0.595).

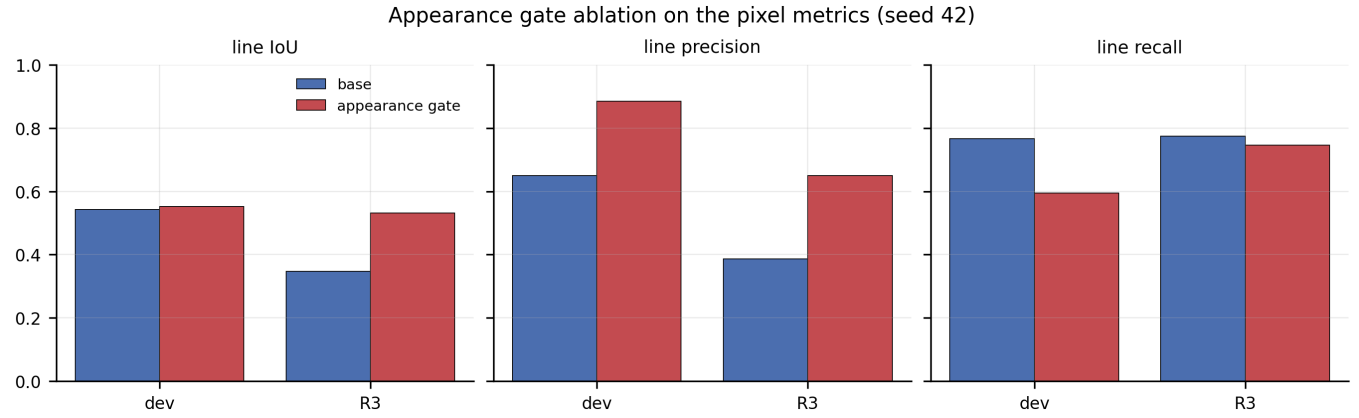


Fig. 3. Appearance-gate ablation on the pixel metrics (line IoU, precision, recall) for both sets, seed 42 — the only switch that trades label recall for precision (Table V).

The appearance predicate is a mask gate, not a truth criterion: its own module documents 13k–68k px of hits on certified line-free frames (bright pavement and bright gravel both satisfy the white branch), which is why it is used to remove candidates rather than to declare where lines are. With that caveat, the instance-level probe records high annotation coverage for the observed limited-class reference instances, with some values below 1.0. This finite sample does not establish complete truth for every surviving candidate. Same-side parallel merging is a separate post-processing factor with a small identity effect in Table V; its radius scan must not be confused with the lateral-scope scan. The two mask thresholds were swept as well; on this pool the negative side is not measurable (no eligible negative frames), so that sweep is reported in the supplement rather than plotted as a zero.

Table V. Post-processing single factors on the frozen development pool, one switch at a time (six seeds each; C0 is the baseline definition, C4 the adopted one). The lateral scope is an identity lever (identity 0.432 → 0.621, all six seeds positive), the appearance gate is a precision lever that pays label recall (0.610 → 0.894 precision, 0.736 → 0.559 recall) and does not raise identity, and the merge is a small identity effect (+0.027).

Configuration	Label precision	Label recall	Paint recall	GT pixels kept	Identity
v7: all three off	0.610	0.736	0.813	0.693	0.432
+ lateral scope 5.5 m	0.610	0.736	0.812	0.693	0.621
+ same-side merge	0.610	0.736	0.813	0.693	0.459
+ appearance gate	0.894	0.559	0.812	0.693	0.409
v8: all three on (adopted)	0.894	0.559	0.812	0.693	0.665

C. Training-composition dose response

4 shows the dose response as it now stands, and the earlier curve did not survive its own check. That curve (base one seed, 4× six, 6× one, 8.5× six: 0.707/0.711/**0.763**/0.701) was measured under an earlier configuration whose identity values do not reproduce under the frozen protocol — the same 6× checkpoint reads 0.763 there and 0.667 here — so it was not comparable with the acceptance numbers and has been replaced. The figure now reports a **paired-seed re-evaluation** on the frozen protocol (seeds 42–47 for every dose whose checkpoints still exist; Table VI): 0× 0.658 (0.631–0.690), 4× 0.661 (0.648–0.674), 6× 0.665 (0.645–0.676). The paired per-seed differences against 0× are +0.003 (4×) and +0.007 (6×), with mixed signs (seed 45 is negative for both), so **the three doses are indistinguishable at six seeds** and the claim that 6× is the peak of the response does not survive this test. The 8.5× arm, whose checkpoints were removed in a storage pass, has a *recorded* six-seed mean of 0.701 — higher than 6×, the opposite of the earlier curve's ordering. The delivered composition remains 6× because it was selected before the one-shot confirmation and changing it would require a newly sealed set; what the paper claims about the composition is now bounded to "no dose effect was demonstrated".

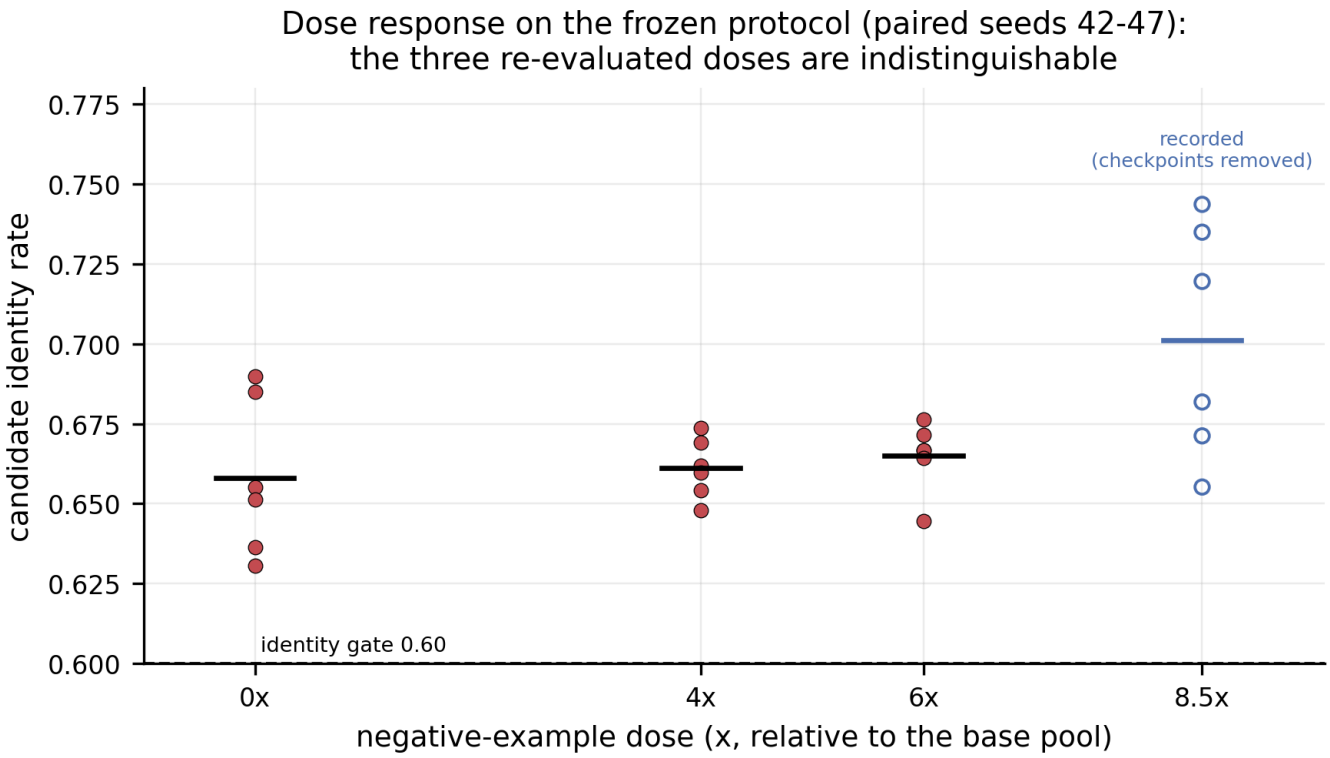


Fig. 4. Dose response on the frozen protocol with paired seeds (42–47): the three re-evaluated doses are indistinguishable, and the recorded 8.5× value sits higher (Table VI).

Line-masked training metrics cannot establish line learning. A separately frozen mechanical check of the unattended learner is documented in the supplement; it is not an additional arm in this dose experiment. The R3 check has since gained a measured mechanism: its training pool composed 15 line-bearing frames with 60 verified line-free frames (line-frame share 0.20), and a pre-registered composition sweep (five paired seeds, four line-frame shares over the same pool family, identical initialization and 60-step budget) shows the development line IoU rising with the line share — the all-line pool beats the diluted composition in 5 of 5 paired seeds (+0.087 ± 0.116) with the smallest seed variance (0.049 vs 0.130–0.145), and precision rises with dose (0.58 → 0.89) at flat recall, while intermediate doses are not separable at this noise level. One diluted-composition seed collapsed to IoU 0.170 against its arm median of 0.439, and the paired comparison now carries an explicit collapse diagnostic. The reading is directional (one pool family, one initialization, a 60-step budget) and no gate changed; what it establishes is that the R3 rejections are attributable, in part, to training-pool composition rather than to the increments being unlearnable. A second engine-verified line-pool family was then admitted after the same projection, coverage and appearance gates: three controlled sites and nine frames, with zero group/content/exposure overlap against the human-revised comparison pool. A five-seed cross-pool comparison matched both arms to nine frames and the same initialization, budget and development evaluation: the engine-verified arm improved line IoU in 5/5 pairs (mean E−H **+0.1003**, SD 0.069; paired verdict `candidate_better`). This is evidence that the second pool family is usable and worth further study, not evidence that engine provenance is universally superior to human revision; pool provenance, scene composition and frame content remain confounded.

A second independent repeat of the cross-pool comparison used new paired seeds (47--51) with the same pools, initialization, 60-step budget and development evaluation. It did **not** reproduce the first-round advantage: engine minus human line-IoU was −0.0574 ± 0.2356, positive in only 2/5 pairs, and the paired verdict was inconclusive. Two engine-arm seeds were flagged by the pre-registered collapse diagnostic; excluding them is shown only as a robustness diagnostic (+0.0972), not substituted for the preregistered result. The combined reading is therefore that the pool-family effect is **not established and is highly seed-/small-pool-sensitive**, not that either provenance is generally superior.

A third, budget-controlled scan kept both pools, the initialization, the five paired seeds and the eight development packages fixed while varying only total optimization steps (30, 60, 120). The engine-minus-human line-IoU difference was −0.0957 ± 0.1354 (1/5 positive) at 30 steps, −0.0622 ± 0.2297 (2/5) at 60, and −0.2139 ± 0.1340 (0/5; `champion_better`) at 120; collapse diagnostics flagged anomalous pairs but did not replace the preregistered verdicts. Thus the second repeat and the budget scan jointly rule out a stable engine-pool advantage: the observed contrast is dominated by small-pool, seed and optimization-budget interactions.

A 240-step independent budget repeat, with the same pools, initialization, seeds and evaluation, made the reversal stronger: engine minus human line-IoU was −0.4106 ± 0.0426, negative in all 5 pairs and `champion_better`; recall was −0.4214 ± 0.0447 in all 5 pairs. Precision was UNKNOWN in all five pairs because no valid positive denominator was present, and was not filled with zero. A third independent human pool (H2) gave a near-zero, inconclusive contrast (engine minus H2 +0.0119 ± 0.0164, positive in 2/5; precision UNKNOWN in 5/5). These results reinforce that pool content, seed and budget interactions dominate this small-pool study; no provenance ordering is established.

A second map was then made admissible by fixing the controlled generator itself: the scenario builder had hardcoded the first map's name into every scenario, the generated road chain sat about 69 m above the terrain, and the camera basis was reconstructed from the wrong direction. After the repair, six controlled collection batches on `west_coast_usa` admitted engine-certified line pools under the same frozen projection, coverage, appearance and provenance gates — 3, 6, 6, 9, 9 and 3 frames across the batches — with every isolated site's reason retained and zero content overlap between batches. Each pool was compared against the same human-revised pool under the identical five-seed protocol (same initialization, 60-step budget, eight development packages): engine minus human line-IoU was −0.1709 ± 0.1497 (3-frame pool), −0.2374 ± 0.1786, −0.2211 ± 0.0614, −0.1072 ± 0.0960, −0.1345 ± 0.0568 (9-frame pools) and +0.1161 ± 0.1195 (3-frame pool) — negative in 25 of 30 pairs, `champion_better` in three of the six comparisons, never `candidate_better`; the last pool was the first whose mean direction was positive (4/5 pairs), yet inconclusive on every metric at five seeds. Admission success therefore does not transfer to learning gains: on the second map, as on the first, the contrast is dominated by pool content, seed and budget interactions — a six-to-nine-frame pool can flip sign wholesale — and no provenance ordering is established. Aggregating all 36 admitted frames from the six batches into one pool did not recover an advantage: the human pool stayed better on all three metrics (`champion_better`, 0/5 positive; line-IoU −0.1918 ± 0.0952, one collapsed seed excluded as −0.1747), so the sixth pool's positive sign is not a pool-size trend. Independently, same-seed reruns of the human arm differ by up to 0.0356 IoU across runs (GPU nondeterminism), a noise floor that every cross-pool comparison carries.

A third map, `east_coast_usa`, exposed a different failure mode and is now also qualified. Its first two controlled batches admitted 0 of 5 sites in each: the generated lane markings were visible in the RGB pass but absent from the annotation pass. A five-scene variable-separation probe falsified elevation, render priority, insertion order and road width as causes, and isolated the mechanism to the annotation pass's alpha-test semantics: the map's shipped `line_white/line_yellow` family scales an opacity factor of 0.7 into base-color alphas of 0.8 and 0.9, so the final alphas (0.56 and 0.63) fall below the material's alpha-test threshold of 255 and every pixel is rejected before the annotation channel is written. An admission rule derived from this mechanism — annotation-writable shipped materials only — re-selected the map's `line_yellow_double` at 0.40 m width, the only setting in a five-width sweep (0.20--0.50 m) that passed the frozen appearance gate in both repeated probe runs (30/30 sampled points showing paint in each; 0.50 m fell back to 0.67/0.63 as the double-stripe gap returned to the truth points). The width change is a pre-registered scene-spec adaptation on this map only; no gate changed. The third batch then admitted 2 of 5 sites (6 frames, `engine_verified`) — the first admission on this map — with coverage 0.9889 and 1.000, calibrated projection medians 0.598 and 0.587 px, and paint agreement 0.9941 at both sites. The three isolated sites are retained with per-instance evidence: one failed all three gates (instance coverage 0.4166/0.2151, projection 10.73 px, paint 0.030), one failed a single instance's coverage (0.4216), and one passed coverage and projection but scored paint agreement 0.0536 — the label-poisoning guard catching annotation marks with no paint in the RGB channel. This pool's first paired comparison against the same human-revised pool under the identical five-seed protocol is negative on every pair: engine minus human line-IoU −0.2684 ± 0.1617 (0/5 positive, `champion_better`), recall −0.2846 ± 0.1645 (0/5 positive, `champion_better`), and precision −0.3281 ± 0.5059 (3/5 positive, `inconclusive`), with two collapsed engine seeds whose exclusion leaves the IoU difference at −0.1690. The human arm reproduced its same-seed values to within 0.0190 IoU of the sixth west-coast run, inside the 0.0356 noise floor that every cross-pool comparison carries. Admission success and learning gain stay separate claims, and on this map the first learning data point is negative: no provenance ordering is established.

The supplement sweeps the lateral scope instead: tightening it from 6.0 m to 3.0 m raises identity (dev 0.582 → 0.713; limited-class 0.581 → 0.758) and removes candidates (477 → 166 and 166 → 127), while role agreement is flat — the lateral scope is an identity lever, not a role lever. Two further single factors were rejected: stochastic weight averaging [18] over the last five epochs rescued no seed and a Tversky loss-weight arm multiplied the false-positive frame rate on the negative side by 4.5 (11.5% → 51.9% of eligible frames, +40.4 percentage points; 329 → 1730 false-positive pixels; maximum component 320 → 808 px), which is why the delivered arm keeps the balanced loss. Across every recorded pixel arm the negative-side readings cluster rather than spread (supplement), which is why the negative-example dose had to be varied instead of a loss-weight knob. Extending the dose response to five metrics (supplement) shows that identity and role move with the dose while coverage and the off-road candidate fraction barely do. The availability cost of the adopted definition is also visible at the pairing stage itself: 3–4 paired frames lost per 117-frame run.

Table VI. Balanced-seed dose comparison on the frozen protocol (seeds 42–47 paired across doses). The three re-evaluated doses are

indistinguishable; the recorded 8.5× value is higher than 6×, so the earlier "6× is the peak" reading does not survive.

Dose	Identity mean	Identity range	Paired diff vs 0×	Role
0× (base)	0.658	0.631–0.690	—	0.874
4×	0.661	0.648–0.674	+0.0030	0.864
6× (delivered)	0.665	0.645–0.676	+0.0069	0.883
8.5× (recorded; checkpoints removed)	0.701	0.655–0.744	—	—

D. One-shot held-out confirmation

The sealed set was read once, by the delivered candidate. Overall label-scope recall is **0.460** and precision **0.763** (IoU 0.403; road IoU 0.633); per group, three of seven groups sit at recall 0.17–0.20 — but not with the same precision: two of them hold precision 0.864 and 0.922 while the third (a27) has precision **0.169** and sends 61% of its predicted line pixels off the road, so it is a group the candidate largely fails rather than a clean high-precision group. One group carries no line ground truth (recall undefined, recorded as such), and the remaining groups range 0.43–0.63. On the negative side 22 of 105 eligible frames carry false line pixels (20.9%), but those pixels are 0.084% of eligible pixels, the largest connected component is 1164 px, and no false candidate appears inside the control region. 5 shows the confirmation. Two properties of this result deserve emphasis. First, it does **not** reproduce the acceptance numbers of Section V-A: the independent set is harder and its group composition differs. Composition is counted two ways and both are recorded: by package, 104 of the 200 frames come from certified line-free packages and 96 from line-bearing ones; by per-frame eligibility, 95 frames carry line truth and 105 are eligible negatives, the difference being frames whose own eligibility differs from their package's label. The delivered candidate's claim is bounded by the numbers reported here rather than by the acceptance table. Second, the confirmation records that the instance-level probes (identity, role, reference coverage) were **not** run on this set; they are UNKNOWN and are not reported as confirmed.

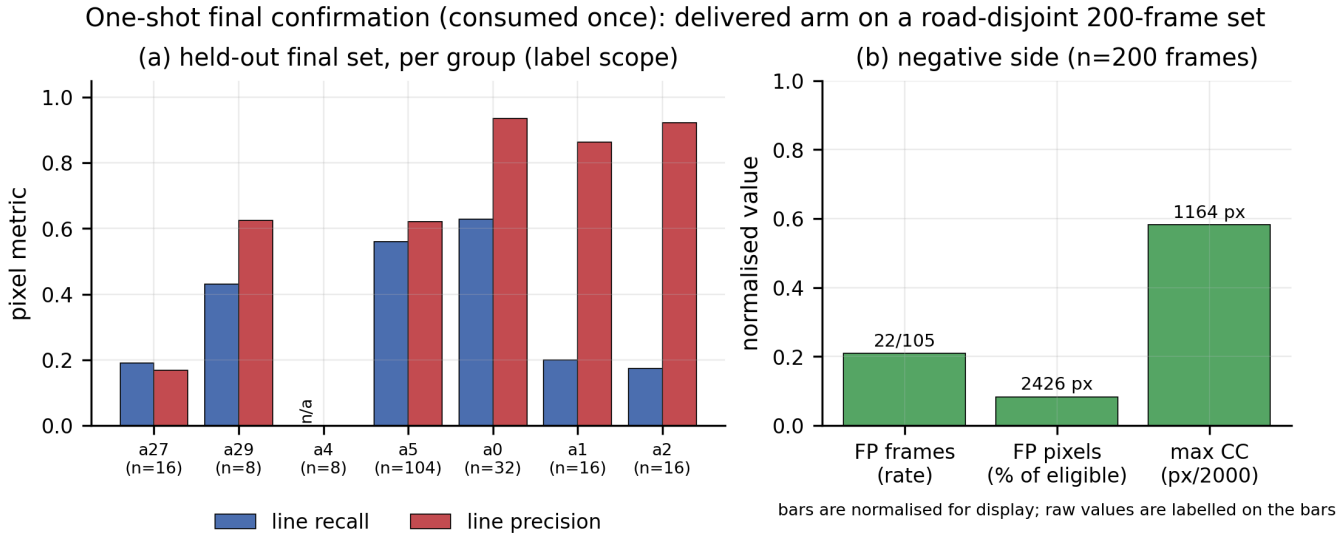


Fig. 5. One-shot final confirmation on the sealed road-disjoint set: per-group metrics (a) and negative-side diagnostics (b).

the supplement states the composition the confirmation ran on (200 frames across seven groups; digest and protocol hash printed in the title so the figure cannot be reattributed to another seal). the supplement is the audit that admitted it: every source package is compared against the 407 frame hashes already used, and the packages that showed overlap were dropped before sealing. Third, and relevant to the gap with Section V-A, the sealed set's truth is **engine-certified** while the acceptance sets are **human-revised**: the two numbers are measured against different truth rungs, not only different data.

E. Closed loop

Definition change in closed loop. With the baseline definition the closed-loop runs show centre crossings of 8, 8, 8 and 13 frames; with the adopted definition no centre crossing was observed in any of the four runs. That is a statement about four runs, not a rate: the same runs travel very little (stall fraction 0.93–1.00, see below), and zero events at near-zero exposure cannot be extrapolated. The adopted definition pays for it with availability: the sensor lane-source rate falls from a median of 0.851 (range 0.231–0.975) to 0.292 (0.097–0.524), a 66% relative drop, and the paired-lane rate from 0.890 (0.809–0.986)

to 0.528 (0.130–0.995). 6 shows the trade-off. Both effects are consistent with the offline finding that the baseline keeps far and non-paint candidates whose pairings pull the car across the lane centre.

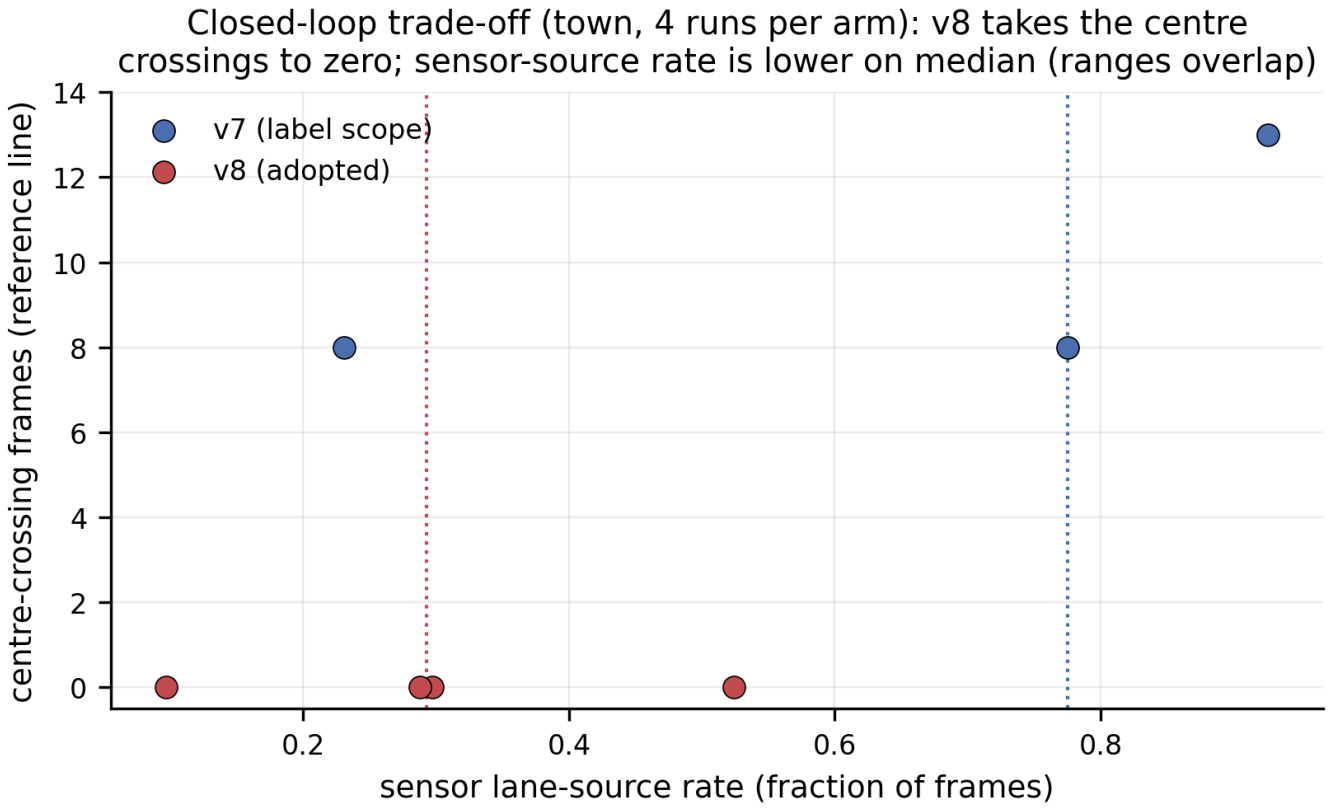


Fig. 6. Closed-loop trade-off: centre crossings versus sensor lane-source rate, four runs per arm.

Acceptance failure and its mechanism. The driving acceptance gate (goal + perception-led lateral mode + strict arbitration, four runs) fails **0/4**: stall fraction 0.93–1.00, travelled 0.9–14.5 m in 120 s, 76.5–88.8 m remaining to goal; the safety-side checks that were measured held (0 collisions, 0 off-road frames, 0 reversing). the supplement shows the per-run hard-target matrix over the recent runs. We then localised the blocker by measurement rather than conjecture. Over 12 town runs, the frames whose reason is "planned vehicle body crosses lane boundary" number 512; in those frames the car is stationary in 96% of cases, the current body is inside the lane in 512/512, and the planned crossing sits at a median of 2.50 m (509/512 within 2–4 m) — i.e. just inside the 4 m hard-stop threshold. The car is parked 0.86 m off the lane centre with its heading ~11° off the lane bearing, so the inflated body sweep of the path the planner actually generated crosses the boundary within ~2.5 m. What the record supports is that the *generated* path is refused, repeatedly, in every frame of that set; it does not establish that no dynamically feasible forward path exists, because the planner's candidate set is not the set of all feasible paths. 7 shows the anatomy.

The audited pose-sweep entry (source/config/exit/cleanup and exposure checked per pose) attempted all nine requested poses: **one of nine was completed and eight were classified** `unplaceable` (`placement_rc=3`, no equal-exposure drive measurement). The completed nominal baseline had 46 frames, 15 accepted-path frames, a perception lane in 26, and a net displacement of 0.56 m; production placement shifted the requested pose by 1.05 m, so it is a `production_alignment` observation rather than a fixed-actual-pose experiment. The remaining six requests were then reattempted with the audited PowerShell/qualification entry (port wait, port rotation, backoff, owned cleanup); all six were again qualified as `unplaceable`. These records establish a **placement-availability failure**, not a causal fixed-pose grid result: the requested pose, the actual post-placement pose and the exposure qualification are not interchangeable. The old raw files remain historical evidence and are not counted as qualified measurements. A fixed-actual-pose experiment would require a placement contract that rejects pose drift before exposure, plus a clean restart between poses. Within the one completed observation, lack of a perception lane frequently precedes path generation; sweep rejection is a separate downstream event and is the principal reason in 8 of the 46 baseline frames. The larger-offset and mountain extensions were then rerun with the explicit `fixed_actual_pose` contract. All four larger-offset poses were rejected before exposure because production placement drifted by 2.57--3.08 m against the 0.10 m tolerance (`placement_rc=4`); all four mountain poses reached the placement deadline without equal-exposure drive measurement (`placement_rc=3`). A re-anchored repeat at the measured actual pose of a +5 m along-

track request after production placement (7.85 m forward, +2.19° heading change; the offset is along the heading, not road-lateral) then **qualified**: position error 0.0209 m, heading error 0.087°, `placement_rc=0`, 26.203 s drive elapsed, 42 frames and 22 accepted-path frames; owned cleanup was verified. A pre-registered start-pose A/B (order O→B→B→O, two repeats per arm) then gave **3/4 qualified** runs, with one O-arm repeat recorded invalid (telemetry exposure too short, preserved as-is). The three qualified runs had near-equal exposure (24.61--24.79 s); the two B-arm repeats differ 2.5× in net travel (3.944 m vs 1.573 m); the single qualified O observation has net travel 3.484 m, inside the B-arm repeat range, while its 20 accepted-path frames sit **below** the B repeats' 26--30 (the two readings are stated separately) — **session-level run-to-run variability exceeds the start-pose difference, and no causal start-pose separation is claimed**; the contrast reinforces our reserve about single closed-loop readings. The two extensions remain placement-contract outcomes, not driving-quality or safety results.

Per-run traces, arm-level distributions and lane-placement distributions are in the supplement. Zero observed collisions, off-pavement frames and reversing in the four formal acceptance runs do not establish safety, particularly under near-stationary operation. The body-footprint centre-crossing criterion failed in all four runs (12--24 frames per run), even when the reference-point crossing metric was low. Availability failures and safety-related body-boundary failures must both be reported.

Four factors run, one proposal refuted before implementation. We pre-registered five driving-layer factors against this blocker; four were implemented and run, and the fifth (pavement re-centring) was refuted by prior measurement before any code was written (Table III). A bounded "hold" window that lets a verified trajectory be reused past its offer age raised motion (reuse 7 → 42 frames; travelled 3.15 → 8.5 m) but worsened the worst-case body-crossing count (29 → 30). Pulling the planned path onto the perceived lane centre engaged in 39 frames and improved the worst case substantially (centre crossings 3 → 0; body crossings 84 → 39) but worsened one right-side item (0 → 2). Removing the obstacle term from the on-pavement gate — see below — doubled travelled distance (6.85 → 14.1 m) and halved the distance to goal (83.5 → 76.6 m) but worsened body crossings (23 → 66) while the car moved twice as far. Swapping in the delivered perception arm produced no separation on path availability (0.179 → 0.200, overlapping ranges) and a worse worst case (36 → 49). All four implemented factors follow the pre-registered maximum rule and are reported as negative; none was adopted, and the fifth remains untested by construction.

The arbitration histogram is the evidence behind the stall fraction: over 2443 settled frames (frames at least 8 s into a run, pooled over the twelve town runs the figure names), "no drivable path" is the largest single reason at **1383 frames**, followed by the planned body-sweep crossing at **512**, which together account for the bulk of the stationary time. A pre-registered production-input replay over the two probe-equipped verification rounds (91 frames) reproduces every recorded safety verdict offline (91/91) and classifies each frame by mechanism: perception-lane loss with a strict refusal of all candidates (23), all 14 planner candidates rejected at the constraint layer (20; cumulative lane-crossing 191 and body-crossing 103 counts), a safety rejection of an already-selected path (28), an accepted path at near-zero speed (19), and accepted motion (1). No rejection happened with a geometrically clear ahead-corridor (0 of 48 rejected frames; nearest corridor intrusion 1.0–5.6 m), so within these rounds the conservative mechanisms responded to recorded near-field evidence rather than mis-stopping on free space. The recorded reason strings are preserved verbatim; a derived per-frame mechanism field separates the conflated `no drivable path` cases.

Table III. The pre-registered driving-layer factors against the deadlock, with mechanism engagement, motion read-outs and the pre-registered verdict. Four were implemented and run; the fifth was refuted by prior measurement before implementation.

Factor	Mechanism engaged?	Travelled (median)	Stall (median)	Worst-case body-cross frames (control → factor)	Verdict
Bounded hold window	yes (reuse 7 → 42 frames)	3.15 → 8.5 m	0.986 → 0.958	29 → 30	rejected
Path re-centring	yes (39 frames)	5.1 → 5.75 m	0.984 → 0.980	84 → 39	rejected (one item worse)
On-pavement gate (obstacle term removed)	yes (availability 0.07–0.52 → 0.36–0.90)	6.85 → 14.1 m	0.966 → 0.950	23 → 66	rejected
Delivered perception arm	no (no separation)	—	0.979 → 0.995	36 → 49	rejected
Pavement re-centring (as proposed)	not implemented	—	—	—	refuted by prior measurement

Why the lane was being dropped. The on-pavement gate revokes the perception lane when the lane centre does not lie on observed drivable surface. Instrumenting the gate's evidence over two diagnostic runs (415 settled frames) shows that of 202 revoked frames, **all 202** were revoked by the obstacle term

(median 6 obstacle cells of ~12 observed samples), while the number of samples that were observed but outside the drivable mask is zero in 192 of them. The obstacle layer is populated from a neural bird's-eye-view head, so on this link the "on-pavement" gate was in effect an obstacle-prediction veto on the lane centre. the supplement shows the evidence and the availability funnel: in the baseline configuration only 116 of 415 settled frames (28%) have a perception lane and only 79 (19%) a planner path.

F. Real-time behaviour and a metric defect

Control cadence. Per-frame telemetry shows the perception-and-planning tick costs 377–432 ms, of which 355 ms is sensor reading (camera ring 139–162 ms, range image 216–239 ms) and 11–13 ms is planning; the neural heads cost 3–5 ms. The renderer itself runs at ~50 fps (20 ms per frame), so the simulation is not the bottleneck: the control loop is paced at 2 Hz of wall clock by design and is in any case tick-bound at ~2.5 Hz. Between ticks a sub-step path re-runs the controllers, up to 15 sub-step iterations per tick, but it is gated on a path having been chosen: in the stall regime (no path) no sub-steps run and commands are issued every ~0.5 s, while in an arm where paths exist 12 sub-steps per frame yield about 31 command emissions per second. These are command-emission rates on the wall clock, not a claim about control bandwidth or about the age of the sensor data each command is computed from. Table IV summarises. The practical consequence is that the perceived "stutter" of the vehicle is a control-cadence artefact of the path-availability blocker, not a rendering or compute failure.

Table IV. Control cadence per configuration: tick cost, sub-step count and command-emission rate (wall clock).

Run	Tick (ms)	Sensor read (ms)	Renderer frame (ms)	Sub-steps per frame	Effective command rate
Baseline acceptance (no path)	432	401	20	0	~2.2 Hz
Lane-gate factor arm	377	355	20	1 (max 5)	~5 Hz
Path-re-centring arm	403	~370	20	12	~31 Hz

T18 sensor replay correctness. After the T17 cadence diagnosis, we froze the source snapshot and ran the T18 replay contract. The first Tech attempt is retained but invalid for equal-exposure comparison: its telemetry span is 18.208 s against the 24 s qualification minimum. A protocol-identical retry qualified. On the first valid warmup capture, the old and current range-processing paths produced **bitwise-equal published geometry and ray-hit outputs in 48 same-input calls**; the best repeated p95 fell from 172.647 ms to 132.420 ms. The scope is deliberately narrow: fresh-tracker single-frame CPU processing, not whole-runtime equivalence, multi-frame tracking, safety truth, actuator execution or a model promotion. Sensor capture time, independent pavement/paint truth, body coverage and actuator acknowledgement remain UNKNOWN; these are not treated as passes. See `T18_SENSOR_REPLAY_CORRECTNESS_RESULT_20261011.md`.

A defect in the safety reading rule. The safety item used to judge every driving factor is a frame *count* of body-centre crossings, and a factor arm can travel twice as far as its control. the supplement quantifies the consequence: the same arm can halve its per-metre crossing rate while showing a larger raw count. This is why, after the first two verdicts, we re-registered a per-distance reading (crossings per 100 m) alongside the raw maximum; the earlier verdicts stand as given, and we flag the rule defect as a methodological result rather than editing them.

VI. Discussion

What generalises. Five artefacts from this programme seem reusable beyond the specific stack. (i) The **provenance ladder with enforced credentials**: naming the truth rung of every label, and refusing to let a command-line string promote one, made "which truth is this number measured against" an answerable question. (ii) The **counting contract**: reporting every ratio with its denominator, and keeping frame-level and instance-level counters separate, removed an entire class of "good number, empty pool" errors and made per-scene failures visible instead of averaged away. (iii) **Protocol versioning with a hash bound to the sealed set**: the mechanism is cheap and it caught a real mislabelling defect; without it, an evaluation produced under one definition would have been attributed to another. (iv) **Appearance certification of generated labels**: requiring that a claimed line be recoverable from the image turned "annotation noise" from a tuning problem into an admission decision. (v) **Pre-registered single-factor A/Bs with a fixed reading rule**, including an explicit refusal to treat unmeasured quantities as passes; this is what allowed four negative results to be reported as results rather than as failures.

What did not work. The scope gap is not closable by engineering alone, and it is worth being precise about why. About 30% of annotated line pixels fail the appearance predicate the adopted definition uses; the predicate reads RGB brightness and chroma only and computes no spatial contrast against the surrounding pavement, so what the measurement establishes is that those pixels do not satisfy *this* predicate. Whether they are visually identifiable to a human, recoverable from context or from another modality, or learnable at all are open questions that this work does not answer — the strong form of the claim ("invisible, therefore unlearnable") is not supported by the measurement. What is supported is the arithmetic: the paint scope retains 69% of annotated line pixels on the development pool, and the retained pixels are recalled at a higher rate than the

removed ones (0.813 versus 0.736 at the same checkpoints), so conditioning the recall on the predicate is not a free improvement and is always reported alongside the full-label reading. Training on the removed pixels remains untested rather than refuted.

The offline gates do not transfer to closed-loop acceptance automatically either: the same perception that clears every offline gate leaves the car unable to move, for reasons (a static body-sweep deadlock) that no offline metric expresses. Finally, availability and safety trade against each other along the definition axis in a way that a single scalar cannot capture: in the four runs we have, the adopted definition was accompanied by no observed centre crossing and by a 66% relative drop in lane-source availability.

Implications for evaluation practice. Three practices would have changed our results materially. Report the *definition* alongside the metric, and make the definition a versioned artefact. Report per-distance or per-opportunity normalisations for safety counts, because motion confounds raw counts. And run one independent, one-shot confirmation whose composition differs from the development pool; our confirmation's negative result is the single most informative measurement in the programme.

Table VII. Configuration of the evaluated arms, read from the delivered candidate's checkpoint metadata. Every arm shares this configuration; the composition arms differ only in training-pool composition and the single-factor arms only in the named switch.

Item	Value
Architecture	`SegUNet`, a lightweight U-Net [19]: three encoder blocks (3x3 convolutions, batch norm, 2x max-pooling), transposed-convolution decoder with skip connections, 1x1 head; channel width factor 1.0
Parameters	834,931 (counted from the delivered checkpoint)
Input	536 x 403 RGB frame
Classes	background, asphalt, line
Class weights	0.147 / 1.0 / 40.0 (line class up-weighted)
Loss	weighted cross-entropy + soft-clDice on the line channel (weight 1.0) + Tversky term [20] (alpha 0.3, beta 0.7, weight 1.0); the rejected arm of Section V-C differs only in the Tversky weights
Optimiser	Adam (beta 0.9/0.999, no weight decay), lr 1e-3
LR schedule	CosineAnnealingLR, T_max 480 steps, stepped per iteration
Batch size	4
Training budget	480 steps (40 per epoch; 12 epochs recorded across resumes)
Sampling	quota sampler over the full pool; per-run validation fraction 0.2 (158 train / 29 validation frames in the delivered arm)
Precision	AMP enabled; deterministic algorithms requested (CUDA cross-entropy has no deterministic implementation, which is recorded rather than hidden)
Initialisation	every arm starts from the same checkpoint (`t14_e1_promotable_20260927/baseline/seed42`, sha16 `b3a4dd5a82dd4096`); that checkpoint's own initialisation lineage is **not** fully recorded (`provenance_complete = False`), so the ancestry above it cannot be reconstructed from the stored metadata
Hardware	NVIDIA RTX 5070; torch 2.12.0.dev+cu128, CUDA 12.8, Python 3.10.11
Matching (frozen)	a candidate matches engine truth when a truth line lies within 0.8 m lateral tolerance; reference availability requires annotation pixels on the candidate's own side; role agreement compares the candidate's left/right role with the engine line's role
Averaging	integer counters accumulated per frame, ratios computed once (never by averaging per-directory ratios); empty denominators are UNKNOWN with a stated reason, never 0
Pools	human-revised anchor 5 directories / 75 frames; development pool 9 deduplicated scenes / 117 frames; limited-class pool 24 frames; sealed road-disjoint held-out set 200 frames

VII. Limitations and Threats to Validity

No instance-level recall is claimed. The counting contract reports coverage, identity and role as conditional ratios over evaluated candidates; a missed marking on a frame where the

perception produced no candidate enters pixel-level recall only. Readers who need instance recall should read the contract's definition (Section III-C) rather than infer one from $|R|/|C|$.

A small human anchor. The human-revised anchor is small (75 training frames) and the human-revised evaluation packs cover 9 packages (117 frames), so scene diversity is limited; the sealed one-shot set is engine-certified rather than human-revised, so its numbers are not directly comparable to the human-revised acceptance numbers. A larger human-revised held-out set would strengthen the definitional argument and is left to future work. The programme deliberately added **no new manual annotation** during the reported rounds, which bounds both the cost and the coverage.

Single simulator, and maps qualified late. The reported perception results come from one simulator. The first controlled `west_coast_usa` batch had zero eligible sites: the generator reported three isolated line sites and three not-applicable no-line sites, with zero raw line pixels at all sites. That failure was later traced to the generator, not to the map: the scenario builder hardcoded the first map's name into every scenario, the generated road chain sat about 69 m above the terrain, and the camera basis was reconstructed from the wrong direction. After these three fixes, six controlled batches admitted engine-certified line pools of 3--9 frames each under the same frozen gates as the first map (Section V-C), so the second map is no longer unqualified — but its pools are small and its cross-pool comparisons are as seed-sensitive as the first map's. The third map (`east_coast_usa`) qualified on its third controlled batch after an annotation-pass alpha-test mechanism was isolated and the scene spec re-selected an annotation-writable material (Section V-C); its pool is 6 frames over 2 sites and its first paired comparison against the human pool is negative (line-IoU -0.2684 ± 0.1617 at five seeds, `champion_better`). A separate native collection there has 803 line-class pixels over 30 forward-camera frames (about 27 per frame); the recorded `utah` forward view has zero over 77 frames. For `east_coast_usa` the competing explanations were resolved — its labels were missing because the annotation pass's alpha test rejected the opacity-scaled shipped line materials, not because of class mapping — but camera pose, frame acquisition, material visibility and class mapping remain competing explanations for the maps not yet probed. The definitional effect's direction, but not its magnitude, is reproduced on a second model (Section V-B). Driving results cover one town link.

Six seeds, and unequal seed counts. Acceptance uses six seeds; composition arms are reported with their seed counts (base 1, 4× 6, 6× 1, 8.5× 6), and the 6× dose point rests on the single seed that was delivered. The dose result is therefore not an effect size: the balanced-seed comparison has been run (Table VI) and finds the three re-evaluable doses indistinguishable, while the recorded 8.5× value is higher than 6×; the 8.5× checkpoints are gone, so that arm cannot be re-evaluated. Frame counts inside a pool are highly correlated, so the seeds are the unit of variation we report and we do not treat pixels as independent trials.

The held-out set is consumed. The one-shot confirmation was performed once, as designed, and the set may not be read again; any future claim requires a newly sealed set. The instance-level probes were not run on it and remain UNKNOWN.

Artefact retention. Per-epoch training snapshots and superseded run directories were removed in a storage pass. Selected verdict files, manifests, telemetry traces, seal records and the delivered checkpoint remain locally. Retained summaries do not recover removed training states or establish complete reconstruction of every figure; the claim manifest records input scopes and remaining gaps.

Driving results are from a research stack. The closed-loop results describe our stack, not automated driving in general; the acceptance gate is a project-internal target, and its thresholds were fixed before the runs and never relaxed.

VIII. Conclusion

We set out to make lane perception for automated driving measurable on a small, explicitly credentialed truth base, and to make its evaluation honest about definitions. A credentialed provenance ladder with four certification gates for its engine rung kept the machine-generated increment trustworthy while the human anchor stayed small and unchanged; a counting contract gave every reported ratio an explicit denominator; and a versioned, hash-bound protocol made the definition of "line" an artefact rather than an assumption. The resulting measurements are informative precisely where they are uncomfortable: the two natural definitions of a line each fail a different gate on the same data, and the disagreement is a third of the annotated pixels; the adopted definition passes all offline gates and records no reference-point centre crossings in four closed-loop runs, at half the availability, while full-body crossing constraints still fail; and the one-shot independent confirmation does not reproduce the development numbers, which we report as the headline of that test. The closed-loop acceptance gate still fails, and we localised the blocker to a measurable deadlock mechanism rather than a tuning shortfall. Every factor, including five negative ones, was pre-registered and judged by rules fixed in advance; we consider that discipline, and the recorded negative results it produced, the main methodological contribution of this work.

Reproducibility and Data Availability

Reported measurements are tied to local recorded artefacts: per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files and sealed-set records. Figure scripts read these inputs; schematics are drawn separately from code definitions. Current fig21 and fig22 have explicit paths and pinned input digests; complete freezing of other historical figure inputs

remains unfinished. Training pools, checkpoints and the sealed set are not distributed with the public code. The availability statement below describes the evidence archive and the audit of removed artefacts.

Nomenclature

Symbol	Meaning
P	eligible frames whose truth confirms a paint line
C	evaluated perception candidates in P frames
C_{out}	candidates that fall on frames outside P
R	candidates in C with an available reference on their own side
M	matched instances, $M \subseteq R$
L	matched candidates with decidable predicted and reference roles, $L \subseteq M$
A	instances whose left/right role agrees, $A \subseteq L$
coverage	R/C (candidate-reference availability)
identity	M/R (candidate identity)
role	A/L (role agreement)
precision, recall	pixel level, reported in label scope and in paint scope

Abbreviation	Meaning
FSD	full self-driving (perception-led driving stack)
BEV	bird's-eye view
IoU	intersection over union
TTC	time to collision
SWA	stochastic weight averaging
PLC	painted-line centring corrector
A/B	alternating control/factor comparison
UNKNOWN	measured quantity unavailable; never read as a pass

Open problems and unfinished work

The programme is deliberately reported with its unfinished parts in the same detail as its results; this section lists them with the evidence that localises each one and the concrete next experiment.

1. The closed-loop acceptance gate is unmet. Four runs of the acceptance configuration fail 0/4 on the benchmark's hard targets (stall fraction 0.93–1.00, travelled 0.9–14.5 m in 120 s, 76.5–88.8 m remaining to goal), while the measured safety items hold (0 collisions, 0 off-road frames, 0 reversing). The blocker is localised to a static deadlock: a stationary car, off-centre by 0.86 m, whose inflated body sweep crosses the lane boundary within a median 2.50 m, so the path the planner generates is refused and the car cannot re-centre (Section V-E, Figs. 7 and the supplement). The record shows that the generated path is refused, not that no feasible path exists; separating the two needs a controlled sweep over frozen poses that stores the perception boundary, the candidate paths, the sweep geometry and the final command for both executable and non-executable cases. The exploratory 3x3 probe has nine output files, but is not qualified as a fixed-actual-pose, equal-exposure experiment (Section V-E). Its single-point pose-set interpretation is withdrawn. *Next experiment*: repair deadline and placement failure handling, record actual post-placement poses, and repeat the invalid conditions under a predeclared pose contract and exposure qualification. Only then judge the three factors that engaged (bounded hold window, path re-centring, lane gate) as a single pre-registered **combined** arm against the best single-factor arm, plus the plan-layer decisions that the programme may not take unilaterally — the planned-crossing hard threshold and the semantics of the on-pavement gate, both of which are safety rules.

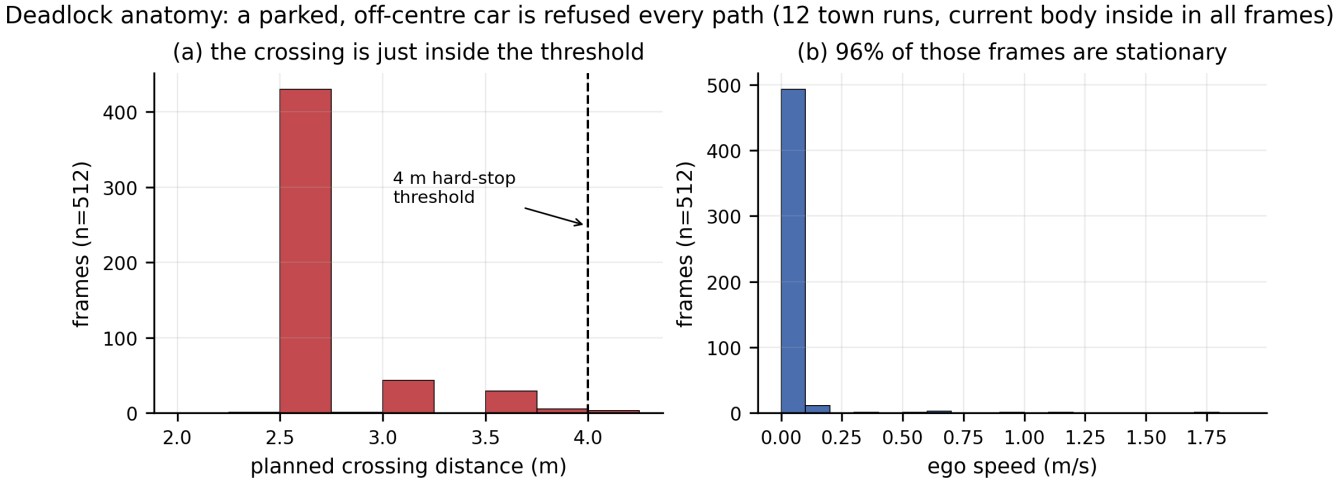


Fig. 7. Deadlock anatomy: the planned crossing sits just inside the 4 m threshold (a) and the car is stationary in 96% of those frames (b).

2. Lateral-reference quality is the next perception-side lever. The planner's path sits a median 0.773 m from its own lane reference while the two layers' references agree exactly (0.0 m gap), and the bounded centring corrector engages only where line evidence exists (39 of 805 settled frames). *Next experiment*:

raise line-evidence availability (a stronger line head, or a fusion that does not depend on the sparse line class), or move the centring upstream so the path is planned on the lane reference rather than corrected after the fact.

3. Dose response: done for three of four doses, and it did not reproduce. The paired-seed comparison has been run (Section V-C, Table VI): 0×/4×/6× are indistinguishable at six seeds. What remains is the 8.5× arm, whose checkpoints were removed in the storage pass, and any dose above 8.5×; both need training runs under the frozen protocol rather than re-evaluation. If a dose effect is to be claimed at all, it needs equal seed counts per dose on more than one map, with the seed as the unit of variation.

4. Control cadence. The perception-and-planning tick costs 377–432 ms, of which 355 ms is sensor reading (camera ring plus range image) and 11–13 ms is planning; the control loop is paced at 2 Hz by design and is tick-bound at ~2.5 Hz. Sub-steps smooth the interval at up to 15 Hz but are gated on a path having been chosen, so the stall regime is exactly where they do not run. *Next experiment*: cheaper sensor reads (ring resolution or round-robin mounts, range reuse under the existing keep-alive) measured as a single factor with both the tick cost and the driving hard targets as read-outs.

5. Specification of safety metrics. The driving factors were judged with a frame-count maximum, and a factor arm can travel twice as far as its control; the supplement quantifies how the same arm can halve its per-metre crossing rate while showing a larger raw count. A per-distance reading has been pre-registered for new rounds; the earlier verdicts stand as recorded, and the rule defect is reported as a methodological result rather than edited away.

6. The observed label/appearance gap. The current appearance predicate excludes about 30% of annotated line pixels. This describes the measured predicate and scope; it does not prove that all local-contrast models must fail or that additional training inevitably causes hallucination. A stronger line head or another sensor modality requires a new controlled comparison. Any change to truth scope, including removing pixels from a recall denominator, must be pre-registered and reported separately from a model improvement.

7. Truth provenance and held-out capacity. The existing human-revised anchor is small (75 training frames; 9 evaluation packages, 117 frames), and the engine-certified one-shot set is now **consumed**. Further confirmation requires a newly sealed set. The next automatic collection must pass independent channel, alignment and provenance checks, retain UNKNOWN for unsupported truth, and freeze development and confirmation data separately. No new manual annotation is planned; results from different truth rungs must remain separate.

8. Second and third maps: qualified after generator repair and material re-selection; both original failure causes resolved. The first controlled `west_coast_usa` batch failed road, line and alignment qualification; the cause was then isolated to the generator (hardcoded map name, ~69 m road-chain/terrain elevation disagreement, wrong camera-basis reconstruction), not to class mapping or the map's materials. After the repair, controlled batches on the second map pass the same frozen projection, coverage, appearance and provenance gates as the first map. The third map (`east_coast_usa`) failed its first two batches because the annotation pass's alpha test rejected the opacity-scaled shipped line materials; re-selecting an annotation-writable material at 0.40 m width — a pre-registered scene-spec change, gates untouched — admitted 2 sites and 6 frames on the third batch. The remaining task is scale: the second-map pools are 3--9 frames per batch, the third-map pool is 2 sites/6 frames and its first paired comparison is negative (line-IoU -0.2684 ± 0.1617, `champion_better`), and paired comparisons are seed-sensitive, so more maps and larger independently qualified pools are needed before any cross-map conclusion.

9. Instance vocabulary for harder geometry. The counting contract's role vocabulary (left/right, near/far) is defined for lane markings on a single carriageway; junctions, 3-D lane estimation, and multi-lane role conventions are undefined, and the identity gate's failure mode on the baseline definition is exactly a role/instance ambiguity rather than a pixel error.

Declarations

Funding. This research received no external funding.

Competing interests. The author declares no competing interests.

Author contributions. The author is the sole author and performed every role: conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualisation, supervision.

Ethics. Not applicable: the work uses a commercial driving simulator and no human subjects. The human annotation referenced in this paper was performed by the author on his own simulator recordings.

Use of AI tools. AI coding assistants (DeepSeek and OpenCode) were used as development tools during the implementation and documentation of this work; all experimental design, measurements, verdicts and conclusions were produced and verified by the author, and every quantitative claim in this paper is regenerated from recorded artefacts by the scripts listed below.

Data and code availability. What is public is the **code and the manuscript sources**: `https://github.com/Qiongkura/BeamNG-autopilot` (branch `fix/round3-hardening-20260921`) contains the evaluation stack, the protocol and counting-contract definitions,

the figure generators, the manuscript Markdown, `build_paper.py` and `references.bib`. What is **not** public is the recorded evidence: the per-run scorecards, per-frame telemetry traces, run manifests, gate verdict files, sealed-set records and checkpoints live under `logs/` in the author's working copy, which is not tracked by that repository (the ignore rules exclude it). Every number in this paper is regenerated from those local artefacts by the scripts named in Appendix A, and `CLAIMS_20261007.md` lists, for each claim, the exact input file, protocol, seed set and scope it rests on, so the mapping can be checked file by file. A public archive of the artefacts is **not yet available**; requests should go to the author. What does exist is the packaged evidence set itself: `scripts/m5_paper_evidence_archive.py` collects the declared numerical evidence, corresponding source and frozen plot inputs into one archive together with a machine-readable manifest of paths, sizes and SHA-256 digests, under `logs/paper_release/` in the author's working copy. That directory is excluded from the public repository by the same rule as the rest of `logs/`. The manifest records the actual file count, missing inputs and digest violations; an incomplete archive fails validation. It provides a checklist for requesting this evidence, but excludes model weights and raw training frames and does not by itself support full training reproduction. Per-epoch training snapshots and superseded run directories removed in a storage pass are listed in a machine-readable audit file (`logs/_cleanup_20261006.txt`), and that removal is why the balanced-seed dose comparison of Section V-C covers three of four doses.

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Appendix A: Recorded artefacts behind each results section

Results section	Artefact(s)
V-A acceptance	`r2_verdict_v7_20261005.json`, `r2_verdict_v8_ADOPTED_20261005.json`
V-B definitional gap	`r2_verdict_v8_ADOPTED_20261005.json` (per-seed label and paint recall)
V-B boundary map	`r2_verdict_v8_paint_{nodilate_mean,dil1b,dil2}_20261005.json`
V-B appearance gate	`dev_pixel_{base,appgate}_20261004.json`, `r3b_pixel_{lat3,appgate}_20261004.json`
V-B scans	`dev_scan_{lateral,parallel}_20261004.json`, `r3b_scan_{lateral,parallel}_20261004.json`
V-B identity scopes	`t16_dual_scope_10seed.json`
V-C dose response	`dev_full_arms_20261005.json`, `dev_full_dose_arms_s4{3..7}.json`
V-C rejected factors	`r2_verdict_swa_20261005.json`, `dev9_beta08_v8d1_20261005.json`
V-D confirmation	`final_set_v5_20261005/seal/{final_set_seal.json,confirmation_base6x-seed42.json,final_set_ledger.jsonl}`
V-D composition audit	`final_pool_audit_20261005.json`, `collect_isolation_20260927.json`
V-E driving	`logs/fsd_benchmark/{scorecard,manifest,town}_*.json` (all runs of every A/B)
V-F timing	`timing_retest_20260927.json`, per-frame telemetry `frame_ms`/`tick_ms` fields
IV pool screening	`t16_r3_density_dev.json`, `t16_r3_pool{,b}_density_scenes.json`, `e1_negative_pool_20260927.json`
IV cost ledger	`gpu_ledger_machine.json`

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